

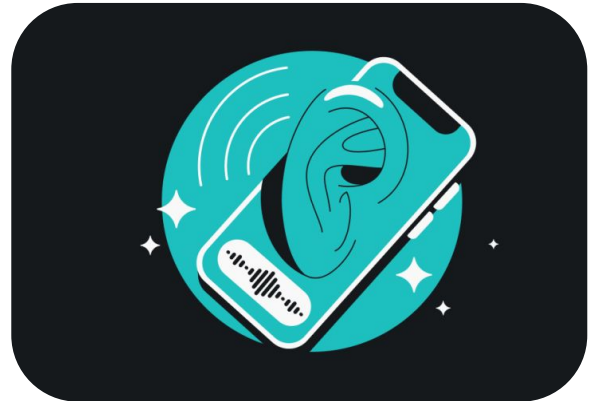
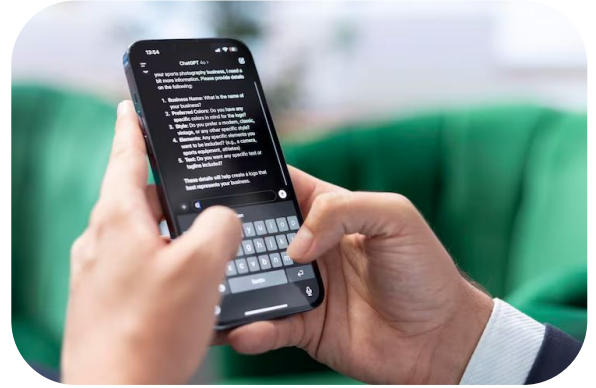
TACT: Touch-free Acoustic Control Triggers for LLMs

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Agenda

- Problem Statement
- TACT and How It Works
- Demo
- Evaluation
- Challenges/Limitations
- Future Steps

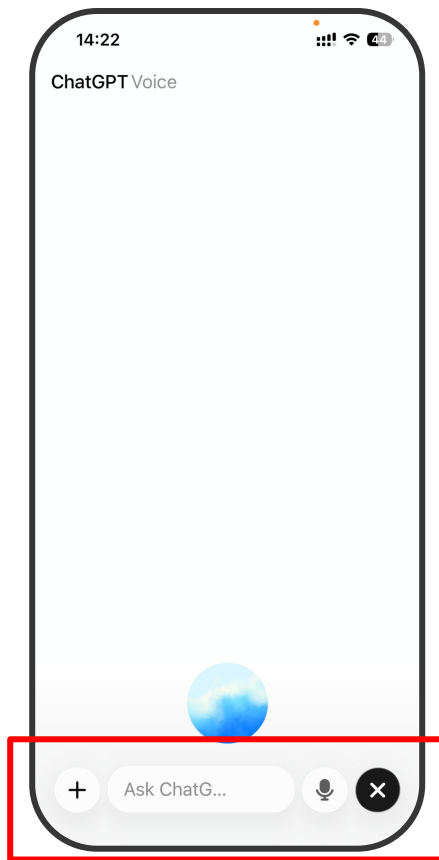
Problem Statement



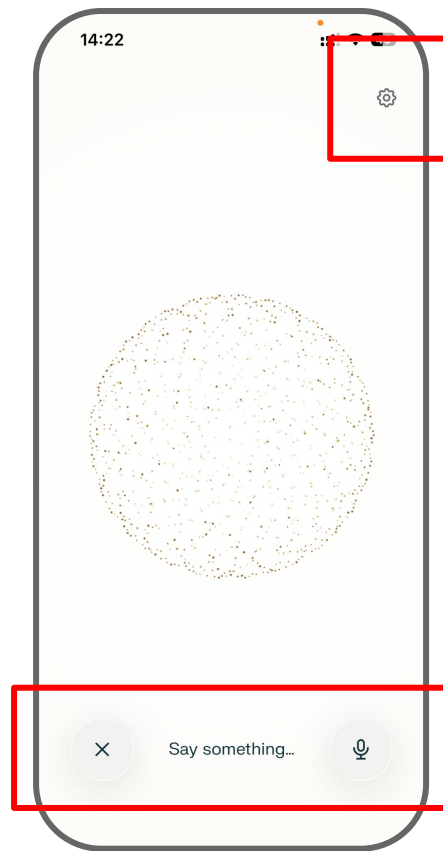
Problem Statement: Rely on Physical Input



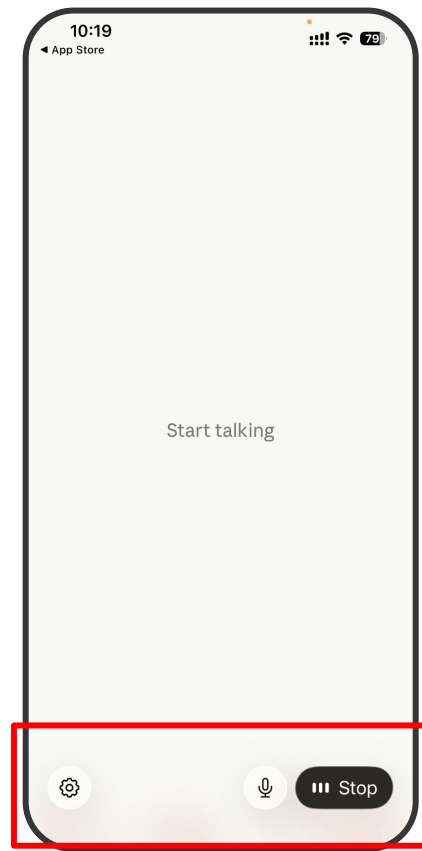
Gemini Live



ChatGPT Voice

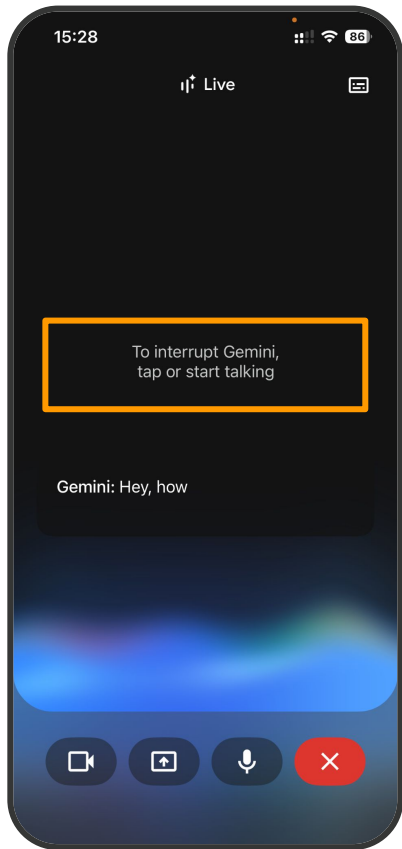


Perlexity Voice

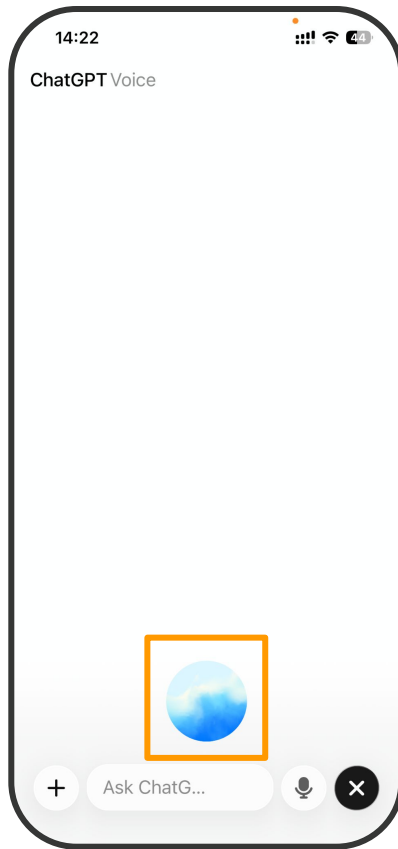


Claude Voice

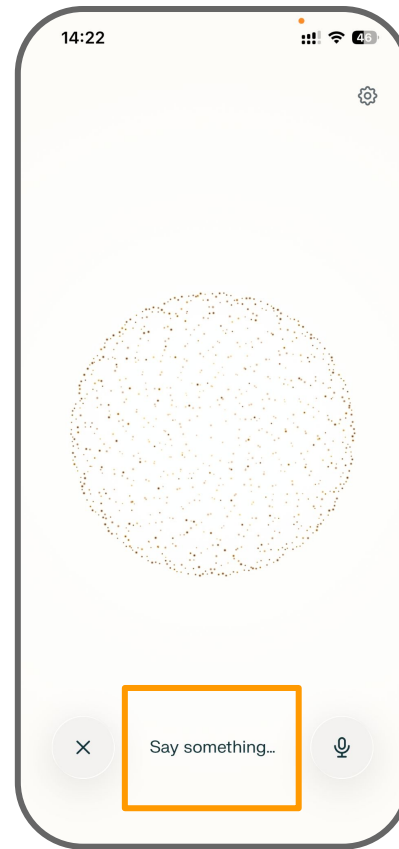
Problem Statement: Rely on Active Listening



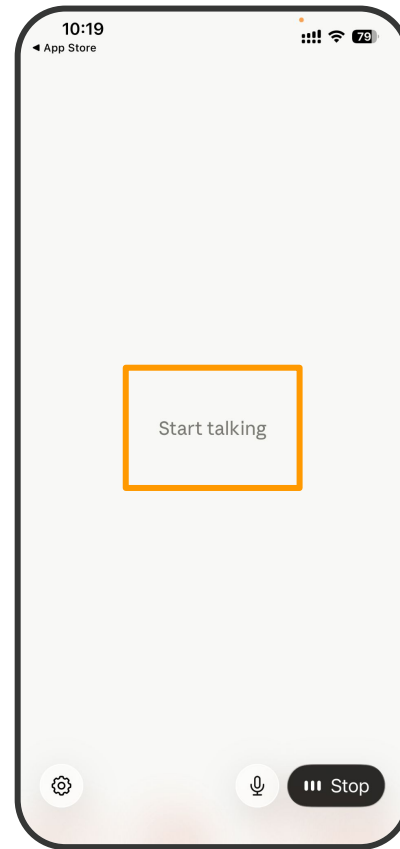
Gemini Live



ChatGPT Voice



Perlexity Voice

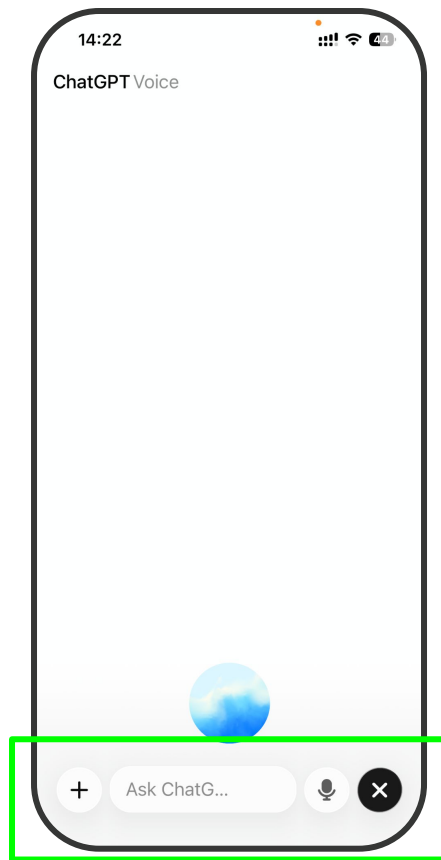


Claude Voice

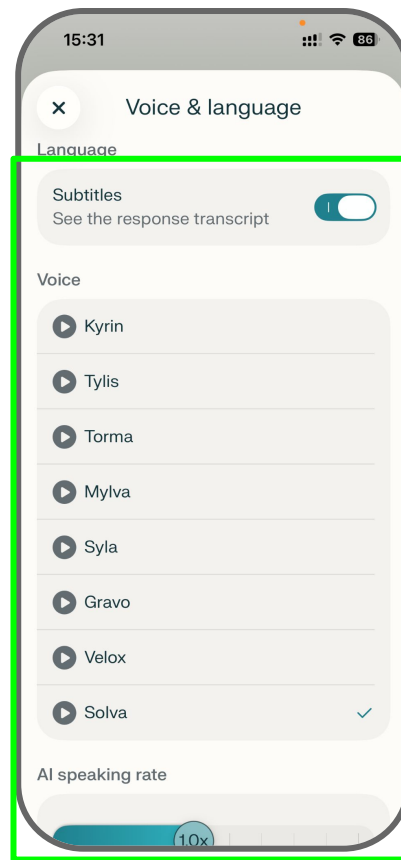
Problem Statement: Limited Settings



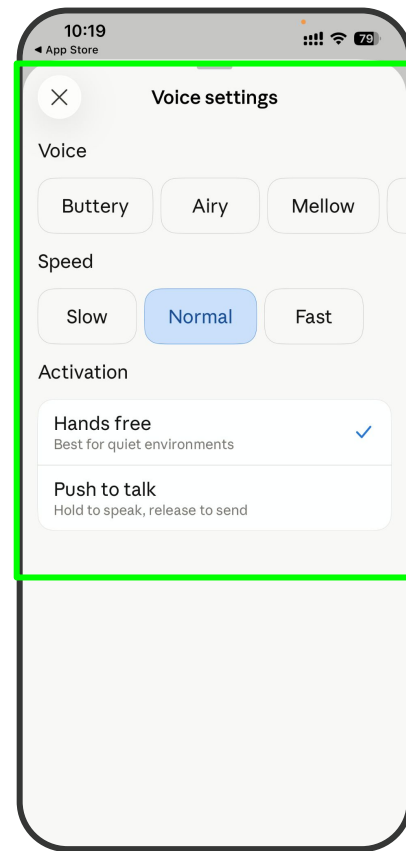
Gemini Live



ChatGPT Voice

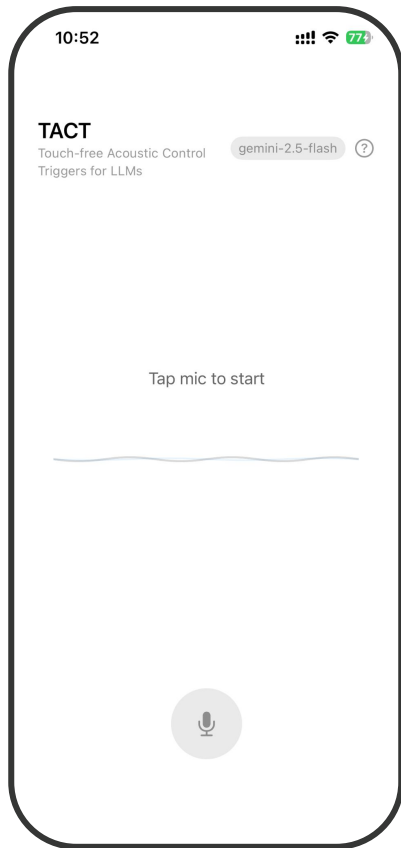


Perplexity Voice

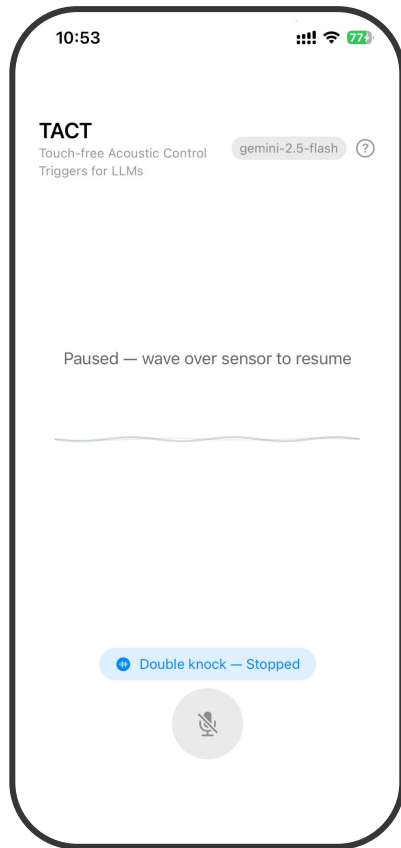


Claude Voice

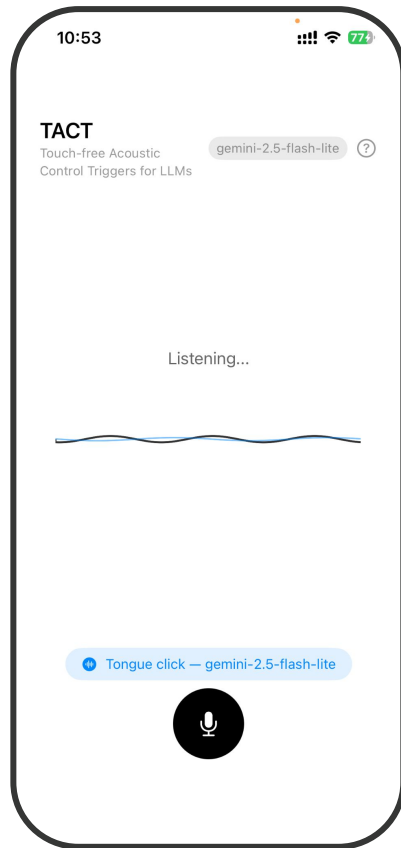
TACT: Touch-free Acoustic Control Triggers for LLMs



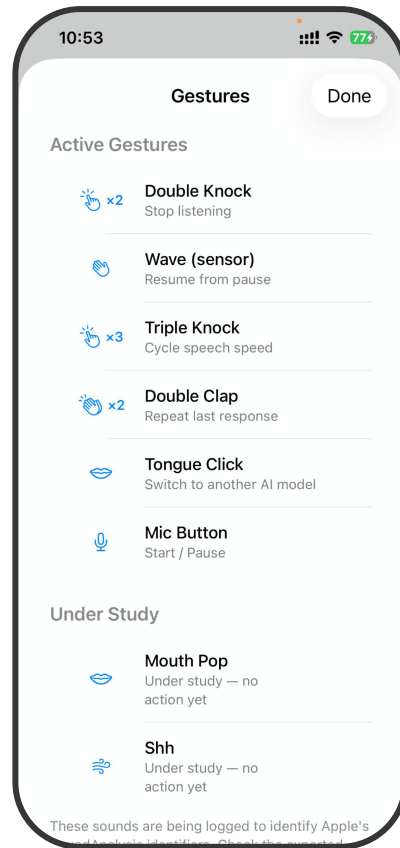
Start Screen



Double Knock Detected



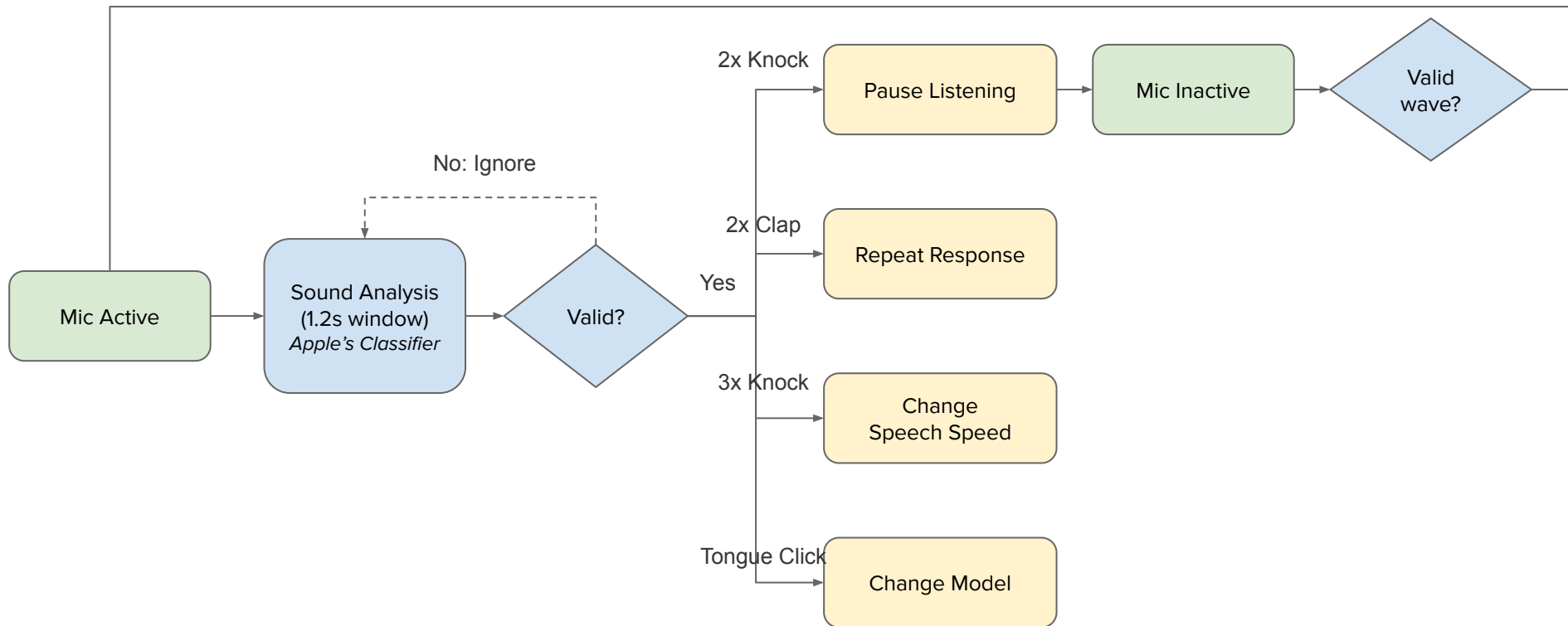
Tongue Click Detected



Guide

How It Works: System Design

Turn on the Mic

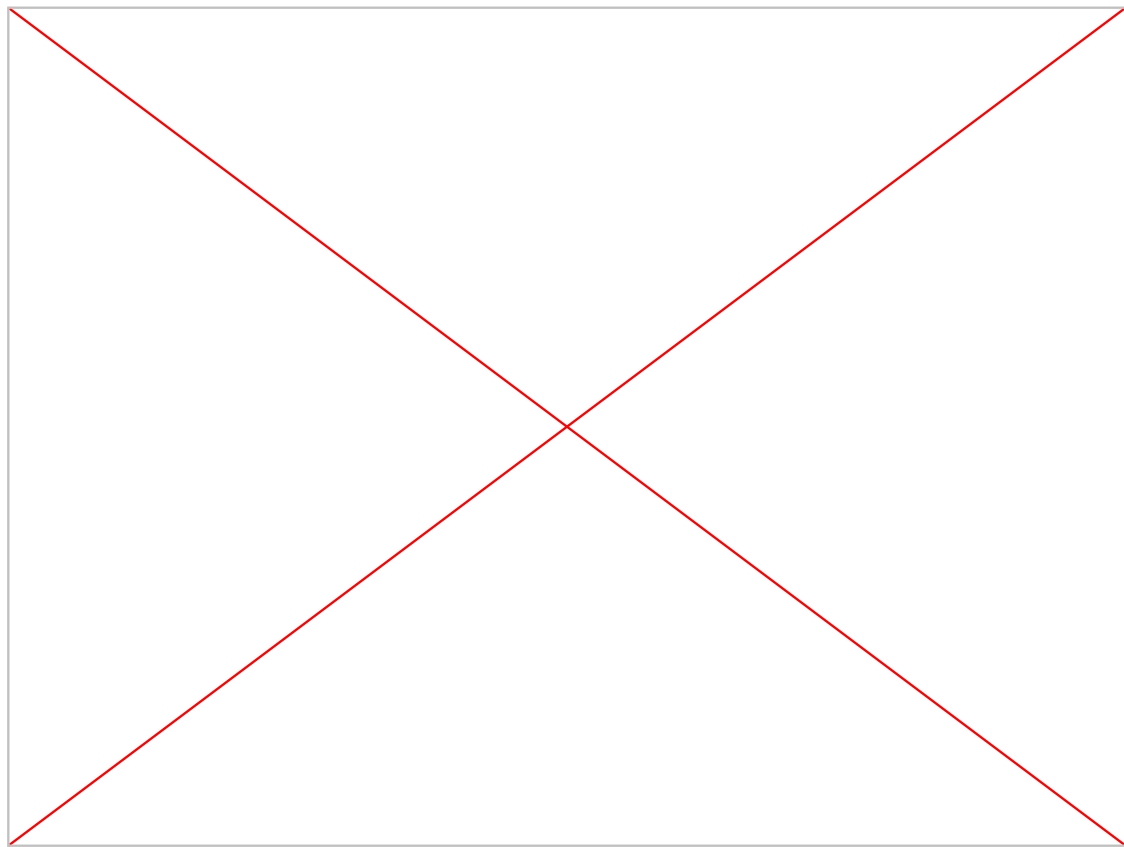


How It Works: Trigger Mapping

Trigger/Gesture	Action	Notes
Double Knock	Pause Listening	
Wave (over sensor)	Resume Listening	Gesture-only with mic off
Triple Knock	Change Speech Speed	Normal → Slow → Fast
Double Clap	Repeat Last Response	
Tongue Click	Switch to Another Model	Gemini Flash 2.5 / Gemini Flash 2.5 Lite

- Tested mouth pop and shhh sounds, but excluded for this implementation due to accuracy.

Demo



Evaluation

Test 1. User Proficiency (Qualitative Task)

[Interactive Scenario]

- **Task:** Writing "Congratulations" in Spanish
- **Action:** Combined **3+ TACT gestures**
- **Stress Test:** Intentional verbal interruptions

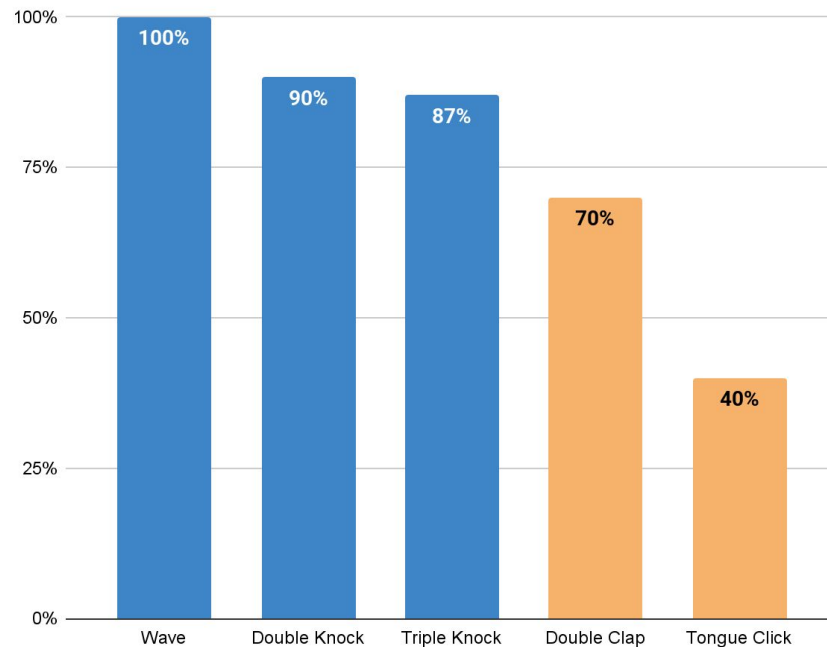
[User Performance]

- **Onboarding:** < 2 mins
- **Task Success:** **100% (5/5)**

[Qualitative Insight]

- **Learning Curve:** *1 participant with distinct tongue click sound initially struggled with gesture calibration but successfully adapted and completed the task.*

Test 2. System Reliability (Quantitative Accuracy)



N=5 participants, n=30 total trials per gesture

Challenges/Limitations

- Distinct Clapping and Tongue Click Sounds
 - Some people cannot make specific sounds.
- Trade-off of recognizing short sounds and preventing False Positive

Future Steps

- Switch to Google YAMNet
 - Can be fine-tuned
- Test more triggers and enable customizations

Thank You



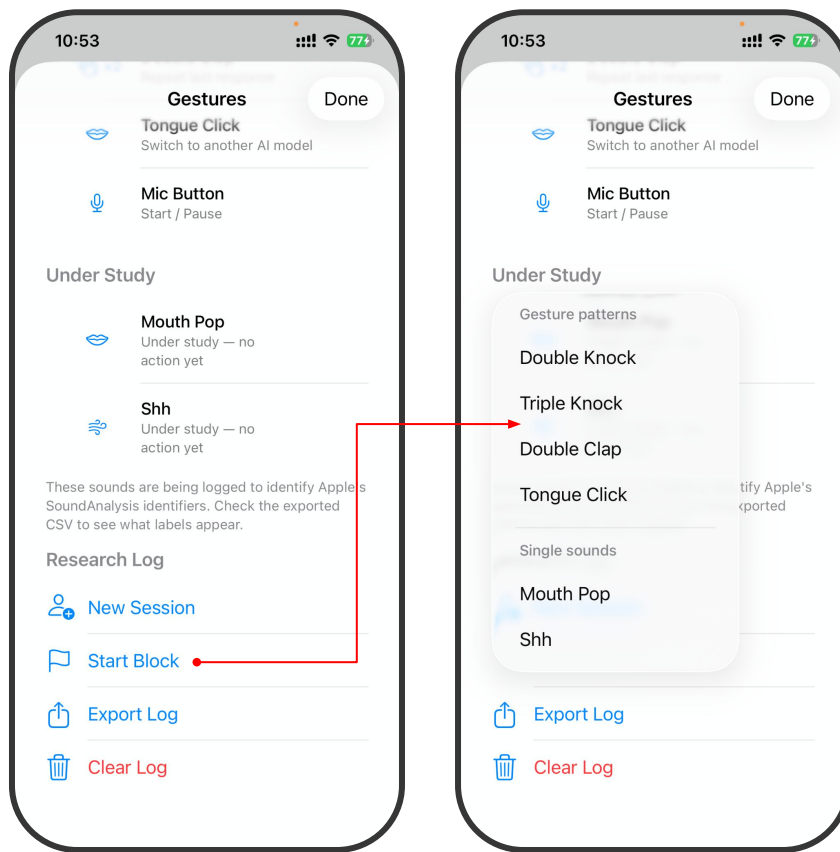
Appendix: Detailed Accuracy Matrix

Gesture	PP1	PP2	PP3	PP4	PP5	Grand Total
Wave	6/6 (100%)	6/6 (100%)	6/6 (100%)	6/6 (100%)	6/6 (100%)	30/30 (100%)
Double Knock	6/6 (100%)	4/6 (66%)	5/6 (83%)	6/6 (100%)	6/6 (100%)	27/30 (90%)
Triple Knock	6/6 (100%)	5/6 (83%)	4/6 (67%)	6/6 (100%)	5/6 (83%)	26/30 (87%)
Double Clap	4/6 (66%)	4/6 (66%)	3/6 (50%)	5/6 (83%)	5/6 (83%)	21/30 (70%)
Tongue Click	2/6 (33%)	0/6 (0%)	4/6 (67%)	0/5 (0%)	6/7 (86%)	12/30 (40%)

Observation Notes:

- **PP2:** Bony hand structure; lower resonance in knocks & difficulty in making tongue click sound.
- **PP3:** Sharp clap sounds.
- **PP4:** Thick/low tongue clicks; frequently misidentified as Knock/Hammer due to spectral similarity.

How to Test and Log



Start Screen

Double Knock Detected

How does the log look like

timestamp	fired_as	top1	c1	top2	c2	top3	c3	top4	c4	top5	c5
2026-04-14T19:28:39Z	knock	knock	1.000	door	0.997	water	0.044	tap	0.024	music	0.015
2026-04-14T19:28:39Z	knock	knock	0.987	door	0.933	water	0.098	wind_noise_microphone	0.053	wind	0.050
2026-04-14T19:28:40Z	knock	knock	1.000	door	0.992	music	0.162	keyboard_musical	0.108	tap	0.097
2026-04-14T19:28:41Z	knock	knock	1.000	door	0.999	tap	0.049	music	0.018	chopping_food	0.007

- Example 1: Knock (2026-04-14T19:28:39Z):
knock=1.000, door=0.997, water=0.044, tap=0.024, music=0.015
→ knockConf = 1.000 >= 0.50 ✓
- Example 2: Clap (2026-04-15T16:36:45Z):
hammer=0.333, clapping=0.283, chopping_wood=0.217, tap=0.145, slap_smack=0.095
→ percussiveScore = hammer(0.333) + clapping(0.283) + chopping(0.217) + tap(0.145) + slap(0.095) = 1.073 >= 0.45 ✓
→ knockConf ≈ 0 < 0.35 ✓
→ clapSignalConf = clapping(0.283) + slap_smack(0.095) = 0.378 >= 0.10 ✓

Apple's SoundAnalysis Classifier

A pre-integrated neural network model optimized for Apple's Neural Engine to classify ~300 acoustic categories.

Pros

- **Native iOS Integration:**
- **On-Device Processing:** works offline without cloud latency.
- **Broad Baseline:** Pre-trained on ~300 diverse sound classes, including common sounds (e.g., Knocks, Claps).
- **Low Resource Usage:** Efficient execution on the Neural Engine, preserving battery life.

Cons

- **"Black Box" Nature:** Undisclosed training data and internal logic make it difficult to debug specific misclassifications.
- **Zero Fine-tuning:** The model is fixed and cannot be retrained or adapted for specific user variances.
- **Missing Niche Classes:** Lacks dedicated labels for specific sounds.
- **Fixed Decision Boundaries:** Difficult to find a "universal threshold."

Sound Categorization Strategy & Threshold

Sound	Strategy	Threshold	Debounce	Notes
Knock	Single label (knock)	knockConf ≥ 0.50	0.5s	Raised from 0.45 to prevent clap misfires. Misses edge cases (0.48) but covers typical 0.6+ scores.
Clap	Composite (8 percussive labels)	percScore ≥ 0.45 + clapSignal ≥ 0.10	0.8s	Increased (0.5s $\rightarrow$ 0.8s) to prevent double-counting. Guard blocked 77% of false positives.
Tongue Click	Composite (tongue+click+tick)	tongueScore ≥ 0.35 + speech < 0.50	0.5s	Lowered (0.40 $\rightarrow$ 0.35) for soft clickers. High per-person variance.
Speech	Single label (speech)	topConf ≥ 0.50	-	System Gate: Suppresses gesture detection while the user is speaking.
Shh	Composite (hiss/shush/noise)	shhScore ≥ 0.3	1.0s	Requires sustained sound ($>1s$). Longest debounce to prevent rapid re-triggering.
Mouth Pop	Single label (snapping)	snapConf ≥ 0.75	0.5s	High threshold (0.75) to cut soft claps/clicks. Risks missing weak pops (<0.75).

- **Analysis Window:** 1 second duration.
- **Step/Overlap:** 250ms hop (0.75 overlap factor).